

[Home](#) > [My Courses](#) > [Portfolio Management](#) >[M5: Kelly and Risk Parity: Optimizing for Growth and Risk](#) > Lesson Notes

Lesson Notes

MODULE 5 | LESSON 2

THE KELLY CRITERION EVOLVES

Reading Time	30 minutes
Prior Knowledge	Traditional portfolio theory and utility theory.
Keywords	behavioral finance, bounded rationality, satisfice, prospect theory, cognitive biases, emotional biases.

The first lesson of this module introduced us to the Kelly criterion and the growth optimal portfolio as alternatives to the Sharpe optimal portfolio with its mean-variance optimization. In this lesson, we develop an even stronger theoretical (which is to say, mathematical) understanding of the Kelly criterion's strategy to maximize terminal wealth (the wealth at the end of the trading period, usually considered some number of years). Having this additional background enables us to fine-tune this approach for even better results, whether the objective is larger wealth (described in the first required reading) or lower risk (described in the second required reading).

1. Fractions and Multiples of the Kelly Criterion

In the last lesson, we read about the two extreme scenarios when it comes to investing (or betting): No matter how low the probability of a loss of our investment (unless that probability is exactly zero), repeatedly investing all of our money will eventually end in ruin, i.e., we will end up with nothing. Such an outcome means we would not have anything left over to try investing again. On the other hand, by not investing any money due to the non-zero probability that we could lose it, we miss out on the opportunity to invest and make more money. Somewhere between these two extremes, there must be a happy medium that balances the risk of ruin and the expected return. The Kelly criterion helps us find this optimal amount to invest.

Again, as we saw in the first lesson, while the Kelly criterion does generally result in greater wealth, this comes at the cost of higher volatility along the way compared to a mean-variance optimized portfolio. For this reason, some more risk-averse investors choose to reduce the volatility inherent to the Kelly strategy by "underbetting": that is, investing only, say, half of the optimal Kelly fraction, which we will refer to as the Half Kelly (as in the reading). Less risk-averse investors, on the other hand, may be happy to double or triple the optimal Kelly fraction (which we will refer to as the

Double Kelly and Triple Kelly). How effective are these different versions of the Full Kelly strategy in relation to various risk and performance measures?

2. Assessing a Portfolio Management Strategy: Many Metrics

The authors of the [first reading](#), Carta and Conversano, provide answers to this question using both Monte Carlo simulation (MCS) and historical data. But to answer the question requires that we enumerate some of these metrics by which to assess a trading strategy. Among others, the authors provide statistics regarding the following:

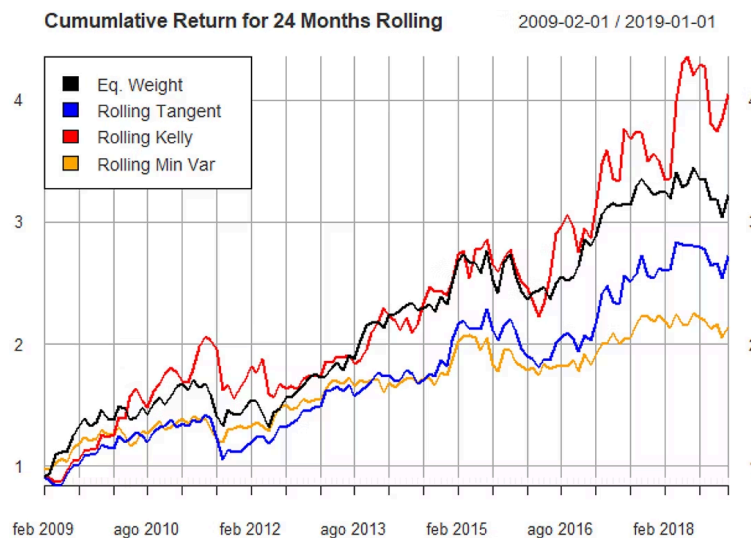
- The mean or the median of the terminal wealth from the many simulations.
- The standard deviation of the terminal wealth from the many simulations.
- The probability that the terminal wealth will be less than the initial wealth.
- The probability that the terminal wealth will be less than half of the initial wealth.
- The probability that the terminal wealth will be less than 10% of the initial wealth.
- The probability that the terminal wealth will be greater than twice the initial wealth.
- The probability that the terminal wealth will be greater than ten times the initial wealth.

The mean is obviously a good measure of the strategy's central tendency but is influenced by the magnitude of outliers: Just one or a few very lucky outcomes might pull up the mean, suggesting that the strategy is usually very positive. The median terminal wealth gives us perhaps a better idea of what we might expect if, say, we have neither good nor bad luck: Extremely high or low outcomes do not factor directly into the median. The rest of the metrics should also make sense as pragmatic determinants of how likely a strategy will meet our goals for wealth increase, with terminal wealth greater than some multiple of what we started with—or alternatively, some small fraction of what we started with.

3. The Time Dimension: Investment Horizon and Lookback Window

Carta and Conversano also relate important insight about the length of the trading period over which the Kelly criterion must be employed in order to reap its benefit of optimal growth. As we discussed in the last lesson, mean-variance optimization (which yields the optimal Sharpe portfolio) is really an optimization for a single trade or a single portfolio without accounting for ongoing trading that involves repeated (if not quite continuous) reinvestment. For that reason, on theoretical grounds, the Kelly criterion might be expected to underperform the optimal Sharpe portfolio if there is only a single trading period or just a few trading periods. Carta and Conversano explore the minimal number of trading periods (just 100 trades, 1,000 trades, and 10,000 trades, and beyond), so we can see how the different versions of Kelly (Half, Full, Double, Triple) perform against the above mentioned metrics given enough time for the geometric compounding from reinvestment to have an impact.

Related to the time component of their analysis is the common sense innovation of using moving averages to compute the mean and covariance matrix for the prospective assets—what Carta and Conversano call “Dynamic Portfolio Optimization.” The figure below shows the performance of the Kelly portfolio using 24 monthly data points in to calculate the relevant inputs, e.g., the means and the covariance matrix, on a monthly basis.

****Figure 1: Out-of-sample Cumulative Returns for 24-months Rolling Average with Monthly Data ****

Source: Carta, Andrea, and Claudio Conversano. "Practical Implementation of the Kelly Criterion: Optimal Growth Rate, Number of Trades, and Rebalancing Frequency for Equity Portfolios." *Frontiers in Applied Mathematics and Statistics*, vol. 6, no. 557050, 2020, pp. 13, <https://www.frontiersin.org/articles/10.3389/fams.2020.577050/full>

Here are some questions for you to ponder as you read the study and then discuss on the forum with your peers:

1. Do Carta and Conversano suggest a simple moving average or an exponential moving average? Can you think of any reason to use one as opposed to the other? Is there any reason not to experiment with both?
2. What is the length of the moving average window suggested by Carta and Conversano? Is that window length theoretically motivated? Is there a reason to try another window length? How might you go about finding the window length that performs best in terms of the metrics listed above? How would a different window length compare to the one they use in terms of the pros and cons they mention?

4. Introducing a Risk Constraint

We saw in the last lesson, as well as in the first required reading by Carta and Conversano, that the optimal growth strategy involves more risk than the mean-variance optimization that yields the Sharpe ratio. In [the second required reading](#), Peterson tackles this issue directly by introducing a risk parameter into the optimization problem. The disadvantage of this approach is the resulting complexity. Given N securities to consider for investment, this approach turns a second-degree polynomial objective function into a fourth-degree polynomial objective function in N dimensions. This might seem daunting, but since you have been exposed to reinforcement learning in your previous coursework, many aspects of the differential evolution algorithm that Peterson employs will be familiar to you.

As you read both papers, pay special attention to how both the risk and the terminal wealth vary in response to manipulating the investigated parameters. Think carefully about which type of strategy you would employ and under what circumstances: Sharpe optimal? Full Kelly? Double Kelly? Half Kelly? Decoupled Kelly? Your answer should likely include not only risk tolerance or risk aversion but also the length of time (i.e., number of trading periods) you have to invest.

5. Conclusion

In this lesson, we reviewed and discussed the theoretical underpinnings of the optimal growth strategy in much more detail compared to the last lesson. We also learned a great deal about the ideal conditions for the Kelly criterion to realize its promise of optimal growth. We took a closer look at the risks involved and several attempts at mitigating these risks, from the simplistic (Half Kelly) to the sophisticated (Decoupled Kelly model with a differential evolution algorithm).

In the next lesson, we continue our review of portfolio construction methods with an emphasis on the interplay between risk and return. Specifically, we look at risk parity, one of the main strategies employed by the largest hedge fund in the world. We will discuss risk parity in its theoretical context (i.e., in relation to the capital asset pricing model) and how it relates to mean-variance optimization and the Kelly criterion.

References

- Carta, Andrea, and Claudio Conversano. "Practical Implementation of the Kelly Criterion: Optimal Growth Rate, Number of Trades, and Rebalancing Frequency for Equity Portfolios." *Frontiers in Applied Mathematics and Statistics*, vol. 6, no. 557050, 2020, pp. 1–15, <https://www.frontiersin.org/articles/10.3389/fams.2020.577050/full>
- Peterson, Zachariah. "The Kelly Criterion in Portfolio Optimization: A Decoupled Problem." *Journal of Investment Strategies*, vol. 7, no. 2, 2018, pp. 1–15, <https://arxiv.org/abs/1710.00431>